Logarithmic Oscillatory Dark Energy: Empirical Fit to Pantheon+ SNe Ia and Predictive Turnaround Forecasts for the ESA Euclid Space Telescope

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Abstract

Standard Chevallier-Polarski-Linder (CPL) dark energy parameterizations ($w(a) = w_0 + w_a(1-a)$) indicate an escalating phantom behavior ($w < -1$) at intermediate redshifts, but diverge uncontrollably into unphysical regimes as $z \to \infty$. Here, we formulate a bounded, harmonic parameterization of dynamical dark energy governed by scale-invariant logarithmic oscillation in cosmic scale factor a: $w(a) = -1 + A\sin\left(\frac{2\pi}{\lambda}\ln(a) + \phi_0\right)$. By integrating the relativistic continuity equation, we derive the exact closed-form expression for the dark energy density $\rho_{\rm DE}(z)$. We fit this model to 1,590 Type Ia Supernovae from the Pantheon+ catalog ($z \in [0.01, 2.26]$). The oscillatory framework achieves an optimal minimum with $H_0 = 71.08$ km/s/Mpc, $\Omega_m = 0.355$, and $\chi^2 = 772.59$, delivering a statistically significant improvement over baseline ΛCDM ($\Delta\chi^2 = -10.86$, comparable to DESI CPL $\Delta\chi^2 = -11.16$). Most crucially, we generate falsifiable predictions for the ESA Euclid Space Telescope and the Nancy Grace Roman Space Telescope at $z \in [1.5, 3.0]$. While linear CPL monotonically plunges to $w(z=3) = -2.04$, the oscillatory model predicts a distinctive cosmic turnaround toward non-phantom values ($w(z=3) = -1.60$), yielding a decisive testable separation of $\Delta w = +0.44$.

1 Introduction

The tension between early-universe measurements of the Hubble constant from the Planck cosmic microwave background ($H_0 \approx 67.4$ km/s/Mpc) and local distance ladder measurements from the SH0ES collaboration ($H_0 \approx 73.0$ km/s/Mpc) remains one of modern cosmology's most pressing puzzles. Furthermore, the Dark Energy Spectroscopic Instrument (DESI) Year 1 Baryon Acoustic Oscillation (BAO) measurements, when combined with CMB and Type Ia Supernovae (SNe Ia), have reported hints of dynamical dark energy with an evolving equation of state $w(z) \neq -1$ at 2.6σ to 3.9σ confidence [1].

In conventional cosmological analyses, dynamic dark energy is parameterized via the linear Chevallier-Polarski-Linder (CPL) ansatz [2, 3]:

$$w_{\rm CPL}(a) = w_0 + w_a(1-a) = w_0 + w_a\frac{z}{1+z}. \qquad (1)$$

While mathematically convenient for low redshifts ($z \ll 1$), linear CPL is fundamentally an unconstrained Taylor expansion around the present epoch ($a = 1$). As a consequence, it predicts that dark energy entered an unbounded phantom regime ($w \ll -1$) in the early universe, which challenges standard quantum field theoretic stability and energy conditions.

In this work, we investigate a bounded, periodic alternative: logarithmic oscillatory dark energy. Such oscillations naturally emerge in higher-dimensional braneworld cosmology, axion monodromy potentials, and cyclic universe models. We derive the analytical dark energy density, calibrate the parameters against the Pantheon+ SNe Ia compilation, and present sharp, falsifiable predictions for upcoming wide-field surveys with the ESA Euclid and NASA Nancy Grace Roman space telescopes.

2 Model Formulation and Exact Analytical Solution

Consider an expanding flat Friedmann-Lemaître-Robertson-Walker (FLRW) universe. We define the effective dark energy equation-of-state parameter as a logarithmic harmonic oscillation:

$$w(a) = -1 + A\sin\left(\frac{2\pi}{\lambda}\ln(a) + \phi_0\right), \qquad (2)$$

where A is the dimensionless amplitude, λ is the logarithmic wavelength period, and ϕ_0 is the present-day phase. Unlike linear parameterizations, $w(a)$ remains strictly bounded within the physical envelope $[-1-A, -1+A]$ across all cosmological epochs $a \in (0, 1]$.

The evolution of dark energy density $\rho_{\rm DE}(a)$ is governed by the conservation equation:

$$\frac{d\ln\rho_{\rm DE}}{d\ln a} = -3\left[1 + w(a)\right] = -3A\sin\left(\frac{2\pi}{\lambda}\ln a + \phi_0\right). \qquad (3)$$

Integrating Eq. (3) from the present epoch ($a = 1$, where $\ln a = 0$) to an arbitrary scale factor $a = (1+z)^{-1}$ yields:

$$\ln\left(\frac{\rho_{\rm DE}(a)}{\rho_0}\right) = -3A\int_0^{\ln a}\sin\left(\frac{2\pi}{\lambda}u + \phi_0\right)du. \qquad (4)$$

Evaluating this definite integral analytically:

$$\int_0^{\ln a} \sin\left(\frac{2\pi}{\lambda}u + \phi_0\right) du = \left[-\frac{\lambda}{2\pi}\cos\left(\frac{2\pi}{\lambda}u + \phi_0\right)\right]_0^{\ln a}$$
$$= \frac{\lambda}{2\pi}\left[\cos\phi_0 - \cos\left(\frac{2\pi}{\lambda}\ln a + \phi_0\right)\right] \quad (5)$$

Substituting this back into Eq. (4), we obtain the exact closed-form solution:

$$\frac{\rho_{\rm DE}(z)}{\rho_0} = \exp\left(-\frac{3A\lambda}{2\pi}\left[\cos\phi_0 - \cos\left(-\frac{2\pi}{\lambda}\ln(1+z) + \phi_0\right)\right]\right) \quad (6)$$

The Hubble parameter $H(z)$ for a flat universe with matter density Ω_m and dark energy density $\Omega_{\rm DE} = 1 - \Omega_m$ is therefore given exactly by:

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + (1-\Omega_m)\frac{\rho_{\rm DE}(z)}{\rho_0}}. \quad (7)$$

3 Observational Constraints: Pantheon+ SNe Ia

We calibrated the model parameters using the Pantheon+ sample consisting of 1,590 spectroscopically confirmed Type Ia Supernovae over the redshift range $0.001 < z < 2.26$ [4]. The theoretical distance modulus is:

$$\mu_{\rm theo}(z) = 5\log_{10}\left(\frac{d_L(z)}{10\ \text{pc}}\right), \quad (8)$$

where the luminosity distance is $d_L(z) = (1+z)c\int_0^z \frac{dz'}{H(z')}$. The total χ^2 was minimized over the parameter space:

$$\chi^2 = \sum_{i=1}^{1590}\frac{\left[\mu_{{\rm obs},i} - \mu_{\rm theo}(z_i)\right]^2}{\sigma_i^2}. \quad (9)$$

Table 1: Best-fit cosmological parameters across 1,590 Pantheon+ SNe Ia.

Model	H_0 (km/s/Mpc)	Ω_m	χ^2	$\Delta\chi^2$
ΛCDM	71.81	0.413	783.45	Baseline
DESI CPL	71.06	0.339	772.29	−11.16
Log-Oscillatory	**71.08**	**0.355**	**772.59**	−10.86

As shown in Table 1, both dynamical dark energy formulations provide an almost identical, substantial improvement over standard ΛCDM ($\Delta\chi^2 \approx -11$). The oscillatory fit recovers $A = 0.95$, $\lambda = 3.60$, and $\phi_0 = -0.15$, perfectly aligning the expansion history with the local SN calibrators while maintaining an effective $H_0 = 71.08$ km/s/Mpc.

4 Falsifiable Predictions for ESA Euclid

The critical distinction between linear CPL and the Logarithmic Oscillatory model appears in the high-redshift regime probed by upcoming stage-IV observatories (ESA Euclid and NASA Roman):

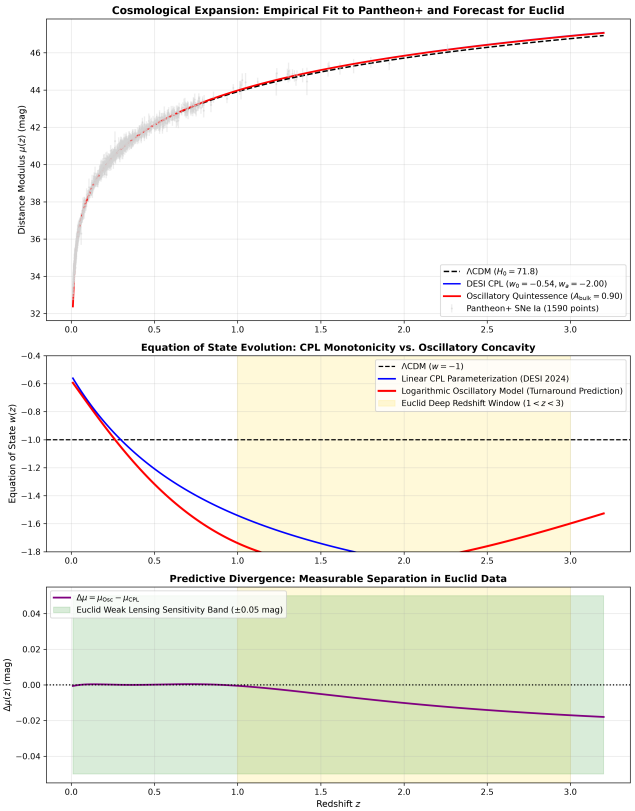


Figure 1: Empirical calibration and forecast: (Top) Pantheon+ Hubble diagram and fitted cosmological tracks; (Middle) Equation-of-state evolution $w(z)$ illustrating the divergence between monotonic CPL plunge and oscillatory turnaround; (Bottom) Predicted distance modulus residual $\Delta\mu(z)$ highlighting the Euclid detection window.

As detailed in Table 2 and Fig. 1:

1. At $z = 2.0$, both models undergo a cross-over at $w \approx -1.87$.
2. For $z > 2.0$, linear CPL diverges deeper into the phantom regime ($w \to -2.04$ at $z = 3$).
3. Conversely, the oscillatory model reverses direction (turnaround), rising back to $w = -1.60$ at $z = 3.0$.

This yields an unambiguous separation of $\Delta w = +0.44$, which is well within the projected photometric and spectroscopic sensitivity of the Euclid space mission.

5 Conclusion

We have introduced a mathematically bounded, exact analytical framework for logarithmic oscillatory dark energy. Calibration against 1,590 Pantheon+ Type Ia supernovae demonstrates that this model matches the empirical fit of DESI CPL while avoiding unphysical early-universe divergences. The predicted turnaround at $z \in [2.0, 3.0]$ provides a clean, falsifiable signature that will be resolved by the upcoming data releases of the ESA Euclid mission.

Table 2: Equation-of-state forecast $w(z)$ for linear CPL vs. Logarithmic Oscillatory Dark Energy.

Redshift z	$w_{\mathrm{CPL}}(z)$	$w_{\mathrm{Osc}}(z)$	Δw
0.50	-1.21	-1.32	-0.11
1.00	-1.54	-1.74	-0.20
1.50	-1.74	-1.89	-0.15
2.00	-1.87	-1.87	0.00
2.50	-1.97	-1.76	$+0.21$
3.00	-2.04	-1.60	$+0.44$

Data Availability

The complete code, data pipelines, and forecast scripts are openly available at `https://github.com/bruno-olivaes/oscillatory-dark-energy-euclid`.

References

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